Soal Labs

TECHNICAL ASSESSMENT

One memo, *every firm*

Build the memo engine for a platform that serves many investment firms: each firm's memo in its own template and rules, edited live in the app, by AI or in PowerPoint, and still right when the data or the template moves.

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01 · THE CHALLENGE

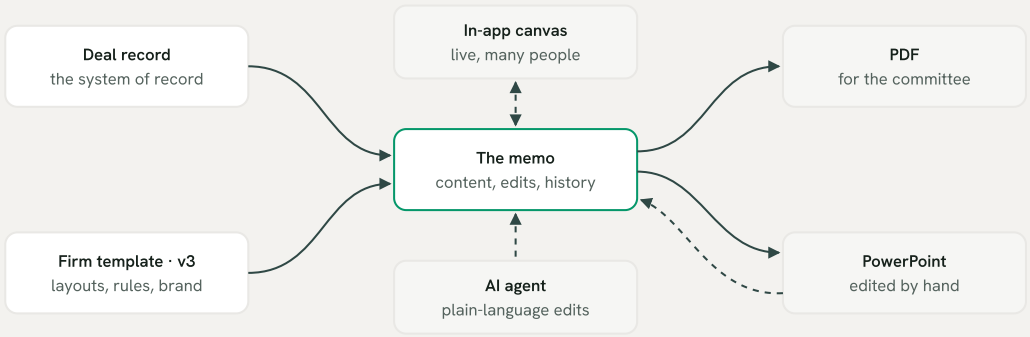
Every firm wants its *own memo*

An investment memo is a deck of eighty to a hundred and twenty slides, and no two firms build it the same way. Different sections in a different order, slides that may never move, a house brand, their own way of writing £22.9m. The memo is where a deal team spends its last week before committee, and every number in it has to match the deal record.

Picture a platform that many investment firms use, where each one wants its own memo without paying for its own code. Software that generates memos meets the same failures every time. Fixed templates overflow, so tables get cut off and footnotes disappear. Teams build the slides they need in PowerPoint and paste them back as pictures, and the numbers on those pictures go stale. Give people a blank canvas and the memo stops looking like the firm's; lock everything and they leave for PowerPoint. Let a model edit the file and it fixes the figure in the text but not the chart beside it.

Build the engine that makes one deal's memo work for any firm: a templating and layout engine, a live in-app editor with a bounded drag-and-drop canvas, custom slides made three ways, AI edits that cannot break the memo, and a round trip through PowerPoint that loses nothing. We are not prescribing an architecture. We assess whether people's work survives every editor and every change to the data and the template; whether you can show where your system breaks and how often; and whether a new firm takes configuration or code. Deciding what to build and what to leave aside is part of the exercise.

ONE RECORD, MANY FIRMS, THREE WAYS TO EDIT



The record holds the facts, the firm's template holds the rules, and the memo holds what people did with both. The canvas, the agent and PowerPoint are three ways to edit the same memo, and every edit comes back through the same checks.

02 · WHAT YOU GET

Two firms, *one deal*

Package A, the firms. Two fictional firms that look at the same company from different seats: Merrow Vale Partners, a growth equity investor, and Corriemoor Credit, a lender. For each, its PowerPoint template, its brand and memo guide, its logo and two past memos. The past memos are real working documents: people broke the template in the ways people do, and built one-off slides by hand that you can use to test upload. Telling the firm's rule from someone's accident is part of the task. A third firm arrives during the build.

Package B, the deal. Tidewell, a fictional UK company selling practice software and payments to veterinary clinics, as a structured deal record: fields with values, units, periods and sources, tables, time series and narrative. It is the system of record your memos render from, and it will change while you build.

03 · WHAT IT MUST DO

Six capabilities, *one memo*

1. **A firm's template lives in the platform.** Layouts and placeholders, a layout grid, which zones are locked, reorderable or free, the brand, components (KPI tiles, tables, charts) and number, date and currency conventions are versioned configuration in your system, not a file. A firm can build its template from scratch in the app, or start from its PowerPoint master and past decks, which a model may read into a draft that a person approves. A deal stays on the template version it started with until someone moves it, and moving it shows what changed and what needs a person.
2. **One memo, every output.** The same memo renders as the in-app preview, a PDF for committee and an editable PowerPoint built natively on the firm's layouts: its own master when it has one, a master generated from its template when it does not. Text is measured in the firm's real fonts; content that runs long continues or splits by the firm's rules, and where it cannot fit, the system says so instead of shipping it.
3. **A live editor in the app.** Template slides stay locked: people edit their content, not their layout. One-off slides get a bounded canvas: drop in images, charts, tables and text, then move and resize them on the firm's grid inside a locked header, footer and sidebar, with figures still bound to the record. Slides and sections reorder by drag where the firm allows it. Every edit is an operation, checked on the server before it lands and shown to everyone in the memo in real time.
4. **Custom slides, three ways in.** Upload a slide from PowerPoint or PDF and get native elements back wherever possible, a flagged picture where not. Build one from scratch on the canvas. Or ask the agent, which builds it from the firm's own components with numbers bound to the record. A one-off slide that works can be promoted into the firm's template once a person approves it.
5. **AI edits are operations.** "Make the title shorter", "add a customer concentration slide", "update the growth rate": plain-language requests become checked changes to the memo, never edits to raw files. A figure that appears in text, a table and a chart changes in all three, and the model never introduces a number the record does not hold.
6. **PowerPoint is an editor, not an exit.** A person edits the exported file and it comes back. Free edits are kept; a number typed over a value from the record is caught and goes to the record or is refused, never silently accepted; broken rules are refused or flagged. When the record changes and the memo is generated again, people's edits survive, and what conflicts is shown, not lost.

Keep the record, each firm's configuration, the memo and every change people made apart, with history, in a real database. Use an off-the-shelf canvas library if you like: we grade the model underneath, not the polish of the interface.

04 · POINTERS

Where the *difficulty* lives

- The platform is the source of truth, not PowerPoint. The canvas, the agent and PowerPoint edit one memo, and a firm with no PowerPoint template has to work just as well.
- Something has to survive export and re-import to say "this is the same slide, the same chart, the same number". Titles, slide order and positions are not identity.
- Decide what each editor may change and what goes back to the record. We will ask first.
- Flexibility is a trap: too much and the memo becomes pasted pictures, too little and people leave for PowerPoint. The firm's rules are the product, and the canvas lives inside them.
- A canvas slide has to leave as native PowerPoint shapes on the firm's layout and come back as the same elements. A picture of a slide is not a slide.
- Two people will edit one slide while the record changes underneath them. Decide what wins.
- A model that edits will be confidently wrong. Decide what checks every change before it lands.

05 · EVALUATION

How you know it *works*

"It looked right when I opened it" is not a result. Build a suite of the edits real people make on the canvas, through the agent and in PowerPoint: drop an image, drag into a locked zone, duplicate a slide, type over a figure, paste a chart as a picture. Run it through repeated round trips with two people editing. Report, for each firm: edits that survive and conflicts surfaced rather than lost; how much of a canvas slide survives PowerPoint and back; overflow on a long-content set you write; AI edits that leave the memo invalid or its linked figures inconsistent; how closely a past memo can be rebuilt; and the time and cost of a render. When the third firm arrives, report how much took configuration and how much took code.

06 · WHAT THIS IS NOT

Three ways to *miss the point*

- **Not PowerPoint in a browser.** The canvas is bounded to the zones a firm leaves free, not a design tool.
- **Not template filling.** Filling placeholders once, with no way back in, skips everything that matters here.
- **Not "let the model edit the file".** Unchecked model edits fail, however good the first demo looks.

07 · FORMAT

A live demo, and the *check-ins* before it

A live demo, not a document submission. We will check in during the build, bring the third firm and put changes in front of you. At the demo, two of us edit the same memo at once, on the canvas and through the agent; we edit a PowerPoint you exported for you to bring back live; and we change the deal record for you to regenerate. We provide API keys; tell us what you need. Where the brief is silent, pick something and defend it. **Stretch goal, only if we raise it:** the same memo as a Word document.